

Load the layer in QGIS

Load the Karnataka boundary layer in QGIS by selecting the load vector layer button as highlighted in Figure 1 (the red square) or on the main menu. Click on **Layer > Add vector layer**. Notice the label *coordinate* at the bottom of the QGIS window. As and when the cursor is moved on the map, the label will show you the X and Y coordinates of the exact location of the cursor. Also, at the bottom-right corner of the QGIS window, you can see EPSG: 4326. This is the code (ID) for the CRS that is assigned to the layer.

Determining the projection of a layer if it is already assigned to the layer

To determine the projection of a layer, right click on the layer under the **Layers** window on the left side of the QGIS window. Next, under the **Properties** window, click on the **Metadata** tab. The CRS assigned to the layer can be seen under the label **Layer Spatial Reference System**. The CRS currently assigned is WGS 84 (red square in Figure 1).

Re-projecting a layer (which already has a projection) to a different projection

A layer for which the projection is already assigned can be changed using the following steps.

Right-click on the layer and click on **Save as**. Next, click on the **Browse** button next to the **Save as** label, name the output file and save it in an appropriate location.

Next, click on the **Browse** button next to the CRS label; at this stage, it shows the name of the projection that the layer is currently assigned to.

Let us choose Google Mercator (EPSG: 900913), which is the CRS used in Google maps. Click **OK**. Tick the option **Add saved file to map**. Again click **OK** to close the **Save vector as...** window. The newly created **shape file** will be automatically added to the QGIS window.

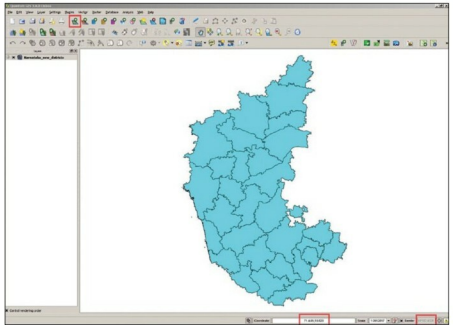


Figure 1: Showing the load vector layer button (red square), projection of the layer (red square) and the X and Y coordinates (red square)

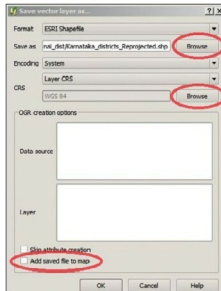


Figure 2: Save as window

Projection transformation (on-the-fly projection)

Once the newly created layer is loaded in QGIS, you can observe that even though the table of contents lists the layer (on the left), the layer is not visible in the map drawing window. This is because the previous layer has a latitude/longitude coordinate system and the newly created layer has a different coordinate system. So the re-projected layer will be located somewhere outside the



Figure 3: Choosing the CRS

current view. To view this layer, right click the layer and click on 'Zoom to layer extent'.

Now the newly created layer can be seen but not the previous one. This error is common while working with layers in different projections.

Ideally, it would be preferable to view the layers on top of each other even if they are assigned different map projections. The option used for this is the on-the-fly projection. To enable this option, follow these steps. First, click on **Settings > Project properties**. Then, click on the box labelled as **Enable on the fly CRS transformation**. Next, select the projection which the current layer